

had to make do with single-layer DVDs or DVD5s, until now. Get ready to cram 8.5 GB of data on just one disk! For the typical workstation backup we mentioned earlier, you can pack it off in only 5 DVDs instead of 10.

Dual-layer burning means you will not need applications such as DVD Shrink to compress DVD9 movies to make them fit on a DVD±R.

We tested a total of eight dual-layer burners in this category: drives from Pioneer, Lite-On, LG, MSI, Sony, ASUS and BenQ. All of them burn dual-layer media at 2.4X. The ASUS DRW-1604P and the Pioneer DVR-108, though go up to 4X.

Features

As was seen earlier, ASUS and Pioneer lead the features pack with the fastest dual-layer burning speed—4X. Strange though it may seem, they also have the lowest CD-R speed at 32X. Every other drive does 40X, barring the Lite-On SHOW-1633S which zooms ahead at 48X.

The ASUS DRW-1604P is a rebagged version of the Pioneer DVR-108, much as the single-layer ASUS DRW-0804P is a

rebagged Pioneer DVR-107, which we haven't tested. Speaking of rebagging, even the Sony DRU-700A is a Lite-On SHOW-832S! The Pioneer we received was a bulky OEM piece—it came sans a box, in a plastic wrap with only one CD containing Sonic RecordNow.

LG receives special mention here for being the only drive to offer DVD-RAM burning at 5X. Only the Pioneer and the ASUS drives support reading DVD-RAMs. Also interestingly, the BenQ DW1600, shipped by default as a single-layer drive can be upgraded to the next model, the DW1620, by a free, official firmware upgrade! We naturally upped the firmware for the tests.

The Sony drive also came with an extra faceplate—a new trend for others to follow? Nokia did it for mobile phones; perhaps Sony will do it for optical drives.

Performance



The Pioneer DVR-108

Demystifying Burners

Not all burners are created equal. Not all of them read and write the same way even if their operational speed is the same. Here are a few strategies that different burners employ:

Constant Linear Velocity (CLV): The advantage of CLV is a constant data-transfer rate. Depending upon the location of the data on an optical disk, the rotational speed of the burner varies: less amount of data is stored near the centre of the disk, so the disk spins fastest to maintain data throughput. As data is read from portions closer to the outside of the disk, its speed slows down, since more data is read per rotation. With CLV, the rotation speed of a disk changes according to the location of the data on the disk.

Zonal Constant Linear Velocity (Z-CLV): Z-CLV uses the CLV mode and further divides a disk into different parts, called zones. A Z-CLV burner starts writing at 6X, and steps up to 8X for the next zone, 12X for the zone after that, and so on. Most high speed burners use this technology.

Constant Angular Velocity (CAV): Quite the opposite of CLV, here the rotational speed remains constant regardless of the physical location of data on the disk. If read in sequence from the centre of the disk right to the periphery, the transfer rate increases smoothly.

The average transfer rate is generally lower than the advertised speed—for example, 16X.

Partial Constant Angular Velocity (P-CAV): P-CAV is a mix of CLV and CAV. Like CAV, the transfer rate goes on increasing until the speed tops out at the rated speed of the burner. From then on, the drive slows down and the data transfer rate remains constant, akin to CLV—by varying the rotational speed of the disk. Using the P-CAV method, burners reach the rated maximum speed faster than burners using CAV, thus achieving a higher average transfer rate.

As far as usage is concerned, Z-CLV is not the best-performing method, because the average performance of a CAV or a P-CAV method is better. For Z-CLV, the time taken to step up the speed to the next zone can add extra time and lower the performance. As an example, the Plextor PX-712A, which uses P-CAV writing, is faster at 12X than the Pioneer DVR-108 at 16X, which uses Z-CLV—the Pioneer burner doesn't step its speed up from 12X to 16X until about 4 GB is burnt. In other words, if you don't burn a full 4.3 GB on the disk using the Pioneer, you'll never reach the 16X zone, essentially making the drive a 12X! If the Plextor was a dual-layered drive, we think it would have taken the Gold away from the Pioneer burner.

The Pioneer and the ASUS were head-to-head in all the performance tests, while also being ahead of the others, thanks to 4X dual-layer burning.

They burned a disk in a little under 27 minutes. The BenQ drive almost caught up, with very competitive reading scores and higher CD scores, but fell back because of its 2.4X dual-layer speed. It took 45 minutes, which was more or less the average score for the rest of the drives.

These three drives—the Pioneer, the ASUS, and the BenQ—all performed well in the DVD+R and DVD-R tests, burning at an impressive 12X. However, the BenQ was the fastest here because of the CAV-writing method as opposed to the Z-CLV method used by the Pioneer and the ASUS.

The LG lagged behind with the lowest scores in both reading and writing—only 4X DVD-R support, and a 12X DVD+R support which actually averaged around 6X. MSI missed out on the DVD-R scores as well. It also revealed a near 4-minute difference between the assorted and the sequential tests.

Price

The Lite-On SOHW-1633S steals the show in this category. Very affordable at Rs 5,750, it is marginally more expensive than its sibling, the SOHW-832S, but offers double the speed for DVD+R media—a very big plus. This is the cheapest 16X dual-layer drive in the market.

The Pioneer and ASUS drives, at Rs 6,300 and Rs 7,250 respectively, offer higher dual-layer performance. LG loses the price battle badly with skimpy performance and Rs 8,000 on the bill, along with Sony, also priced at Rs 7,999—for the extra faceplate? Thanks, but no thanks...

Conclusion

A 16X drive with 4X dual-layer burning for Rs 6,300 is simply unbeatable. Yes, the Pioneer DVR-108 is an OEM piece that is supplied to PC manufacturers such as HP to sell with their PCs, and for that reason, it comes with no media or manual, but we have yet to see those facts make a notable difference to the market. The drive works well, and is cheap. It wins Gold with an overall score of 72.77, and makes for a great choice for speedy backups.